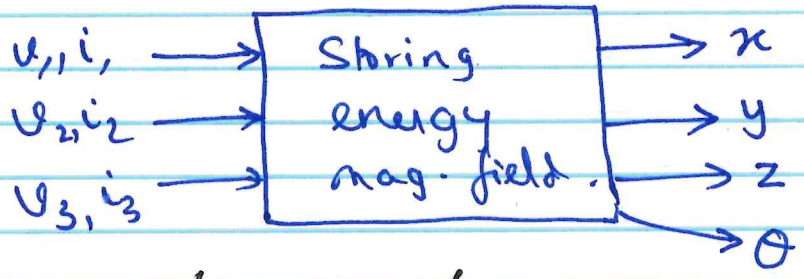
Lecture 7

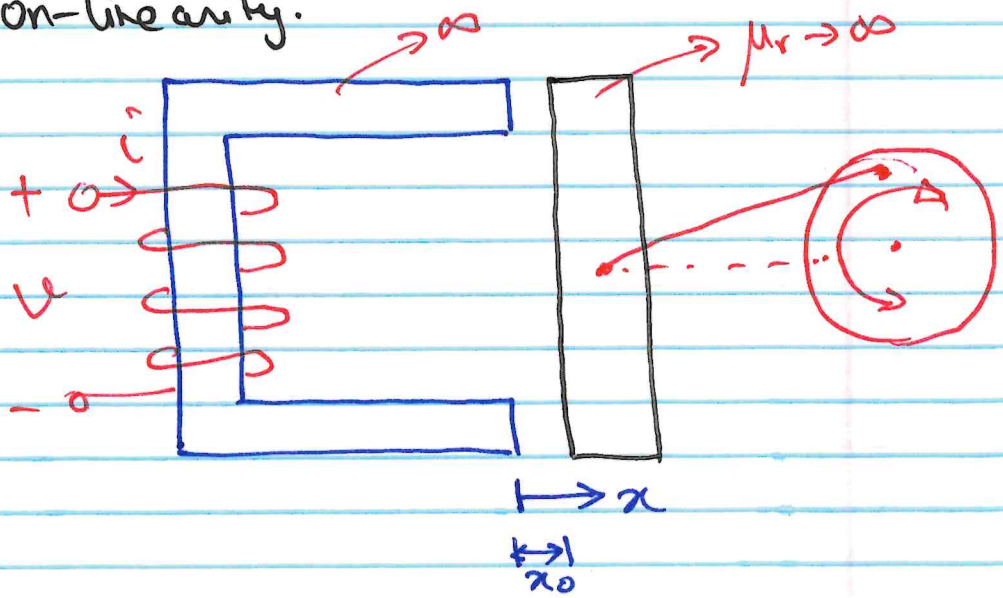
* Electrical ports → one
  multiple
  - energy
  - coenergy
  - forces

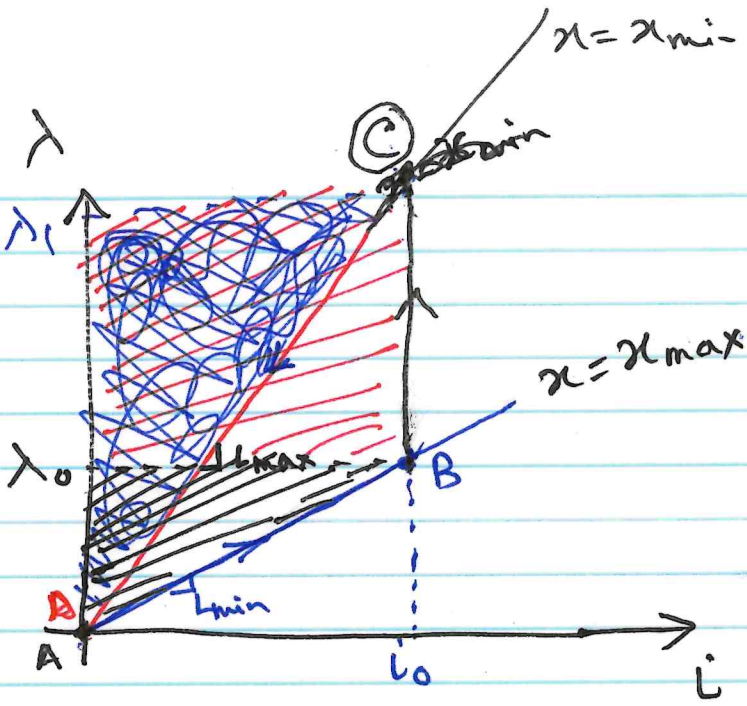
* ~~Multiport mechanical system~~
* Multiple mechanical port.



* EFE / EFM / Cycles ←

* Non-linearity.





$\lambda_0 = L_{min}\, i_0$

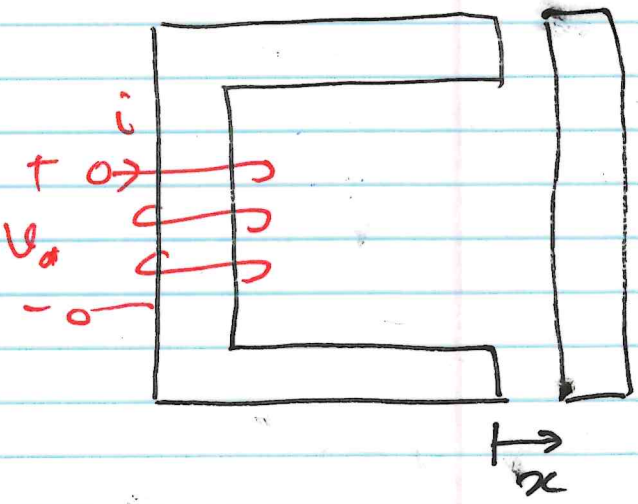
$x = x_{max}$, $i = 0$

@ A

$\omega_m = 0$

EFE = 0

EFM = 0



$x = x_{max}$, $i : 0 \rightarrow i_0$

$L_{max} \leftarrow x = x_{min}$

$L_{min} \leftarrow x = x_{max}$

$$EFE = \int_0^{\lambda_0} i\, d\lambda$$

EFM = 0

$$\omega_m = \int_0^{\lambda_0} \frac{\lambda}{L_{min}}\, d\lambda = \frac{1}{L_{min}} \frac{\lambda_0^2}{2} = \frac{1}{2} L_{min}\, i_0^2 = \omega_m(B)$$

$$d\omega_m = \boxed{i\, d\lambda} - \boxed{f_e\, dx}$$

$d\omega_m$ = EFE + EFM

i : @ $i = i_0$ $\quad x : x_{max} \to x_{min}$

$$EFE = \int_{\lambda_0}^{\lambda_1} i\,d\lambda = i_0(\lambda_1 - \lambda_0)$$

$$EFM = -\int_{x_{max}}^{x_{min}} f_e\,dx$$

$$\omega_m(C) =$$

$$\boxed{\Delta\omega_m = EFE + EFM}$$

$$\omega_m(C) - \omega_m(B) = EFE + EFM$$

$B \to C$ $\quad B \to C$

$$\frac{1}{2}L_{max}\,i_0^2 - \frac{1}{2}L_{min}\,i_0^2 = i_0(\lambda_1 - \lambda_0) + EFM$$

$B \to C$

i : $i_0 \to 0$ $\quad x = x_{min}$

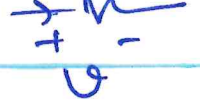
$C \to D$

$$EFM = -\int_0 f_e \cdot dx = 0$$

$$EFE = \int_{\lambda_1} i\,d\lambda = -\frac{1}{2}L_{max}\,i_0^2$$

$i = 0$ $\quad D \to A$ $\quad EFM = 0$ $\quad EFE = 0$

$x : x_{min} \to x_{max}$ $\quad \omega_m = 0$



$$\int_0^t v\,i\,dt'$$

$$\int_0^t i^2 R\,dt$$

$$\Delta W_m = i\,d\lambda - f_e\,dx - i^2 R\,dt$$